Klarify: Kinyarwanda TTS Dataset Review and Verification Platform

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GitHub: https://github.com/Yves-Developer/yves_20251SEN202_klarify

1. Problem Statement

Text-to-speech (TTS) systems require large amounts of accurately transcribed audio. For Kinyarwanda, such data is scarce (Rutunda et al., 2023), and existing datasets contain spelling, punctuation and transcription errors. Crowdsourced transcription is low in cost but introduces errors that require review (Novotney & Callison-Burch, 2010). No structured process exists to correct and verify Kinyarwanda TTS data. The need is clear: 21.2% of Rwandans aged 15 and over cannot read, and 54.0% can read only in Kinyarwanda (National Institute of Statistics of Rwanda [NISR], 2023).

2. Proposed Solution

Klarify is a platform built on an Oracle database. Native speakers listen to audio clips and correct the matching Kinyarwanda transcripts, and peer reviewers verify each correction. PL/SQL procedures, triggers and auditing enforce the review rules. The result is a verified dataset ready for TTS model training. Existing platforms such as Common Voice validate recordings by vote but do not allow reviewers to correct the text (Mozilla Foundation, n.d.).

Main objective: To develop an Oracle PL/SQL platform that produces a verified Kinyarwanda TTS dataset.

3. Data Source

The project uses the *Kinyarwanda TTS dataset* by mbazaNLP (2023), available on Hugging Face. It contains 3,992 studio recordings (about 6 hours) by one speaker, with matching Kinyarwanda text. It is licensed under CC BY 4.0, which allows educational use, and contains no personal information. Some recordings contain reverberation, which makes review necessary.

4. Planned Tables

Table	Purpose
USERS	Contributors and reviewers
CLIPS	Audio clip details and status
TRANSCRIPTS	Current transcript of each clip
TRANSCRIPT_VERSIONS	History of all transcript edits
REVIEWS	Reviewer decisions
FLAGS	Reports of poor audio or text
ERROR_TYPES	Categories of errors (spelling, punctuation, mismatch, noise)
AUDIT_LOG	Record of all changes and denied actions

5. Planned PL/SQL Features

- **Procedures:** submit_correction saves a new transcript version; review_clip records a reviewer decision.

- **Functions:** get_approval_count returns the number of approvals; normalize_text corrects spacing and punctuation.
- **Cursors:** process pending clips for reviewer assignment and dataset export.
- **Package:** pkg_klarify groups the correction, review and export logic.
- **Triggers:** approve a clip after two reviewers agree; prevent users from reviewing their own work; write every change to AUDIT_LOG.
- **Exceptions:** handle self-review, edits to approved clips and invalid status changes.

6. Innovation

Type: User interface (custom web application). A Next.js web application connects to the Oracle database through node-oracledb (Oracle, n.d.). Contributors play audio clips and correct transcripts, and reviewers approve or reject them. Each action calls a PL/SQL procedure, so all business rules remain in the database. A dashboard shows review progress and common error types.

7. References

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